

DRAFT — DBA-ES-GN-FC6: Coal Generation

Coal-Fired Generation

Design Basis Accident — Military Action

DBA-ES-GN-FC6 — Facility Class Document

Issued Under: DBA-ES Electric Sector Framework

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Preface

The coal-fired power plant is a diminishing but indispensable asset. The United States has retired more than half of its coal-generating capacity in the past fifteen years, driven by natural gas economics, renewable cost reduction, and environmental regulation. What remains is a fleet selected by geography, grid necessity, and capital economics rather than by any systematic security analysis. The units that have survived are disproportionately the ones that the grid cannot easily do without — large baseload units in transmission-constrained regions, must-run plants designated by RTOs and ISOs for voltage support or frequency response, and facilities whose steam turbines provide the synchronous inertia that inverter-based resources do not. The adversary who understands grid operations does not need to know the energy policy history of the U.S. coal fleet. They need to know which coal plants the system cannot replace quickly when they are gone.

This document establishes the Design Basis Accident scenarios for deliberate military action against large coal-fired generating facilities that meet the FC6 criticality threshold. The facility class encompasses coal plants above 750 MW individual unit nameplate rating and any coal unit designated as must-run or transmission-essential by the applicable RTO or ISO, together with the on-site coal handling and storage infrastructure and the coal combustion residual management systems that are integral to plant operations. The primary reference environments are the Appalachian and Midcontinent coal corridors, where transmission-constrained load pockets depend on local coal generation for voltage support that remote renewable resources cannot provide.

FC6 carries three analytical contributions unique in the DBA-ES series. The first is the fuel supply dependency: a coal plant's Design Basis must account for the coal delivery infrastructure — the rail and barge supply chain — as a consequence chain component in its own right. No other DBA-ES facility class has a fuel supply chain that can be interdicted as an alternative attack pathway to physical destruction of the generation equipment. The second is the RT_HVLLC governing metric applied to thermal generation: the boiler pressure parts and steam turbine components whose destruction produces the longest recovery timeline in the generation fleet — 18 to 36 months — and whose lead times have no parallel in any other DBA-ES generation type except large hydro. The third is the coal combustion residual consequence chain: an attack that breaches a coal ash impoundment produces an environmental cascade that extends into the water sector, connecting the electric sector consequence chain to the DBA-W series through the same compound-consequence mechanism identified in DBA-ES-HY-FC1 for dam failure.

The evidentiary anchors for FC6 are the 2008 Kingston Fossil Plant CCR impoundment failure — the largest industrial spill in U.S. history at that time, covering approximately 300 acres with 1.1 billion gallons of coal ash slurry — for the environmental consequence chain; the pattern of

railroad disruptions during extreme weather events for the fuel supply Design Basis; and the Ukrainian experience of deliberate infrastructure targeting at thermal power plants as the military-action evidentiary anchor for the kinetic attack scenario.

1. Introduction

1.1 Purpose

This document establishes the Design Basis Accident — Military Action scenarios for the coal-fired generation facility class. Its purpose is to define the bounding threat event that coal plant security planning, resilience investments, and recovery architecture must address. As with all DBA-MA documents, this document does not prescribe specific protective measures at individual facilities. It defines the consequence envelope within which protective measures must perform.

A coal plant security program designed against conventional physical threats — perimeter intrusion, sabotage of accessible equipment — will fail when a sophisticated adversary applies the intentionality and precision characteristics of military action to select the boiler and turbine components whose destruction produces the longest recovery timeline, combines that kinetic attack with a SCADA-based fuel management disruption that depletes the on-site coal stockpile before the attack executes, and times the compound event to coincide with a transmission-constrained system condition where the affected imports cannot replace plant's generation. The DBA-MA approach defines this compound, timed, consequence-maximizing attack as the governing design event.

1.2 Scope and Applicability

This document addresses coal-fired generating facilities meeting either of two applicability criteria: individual generating unit nameplate capacity at or above 750 MW, or designation as a must-run or transmission-essential unit by the applicable RTO or ISO, regardless of nameplate rating. The 750 MW threshold reflects the consequence-chain organizing principle of the DBA-ES series. Below this threshold, the loss of a single coal unit is typically manageable through normal N-1 planning without producing Tier 2 grid-level reliability consequences. Above it, or where transmission-essentiality applies regardless of size, the loss of a single unit can produce Tier 2 consequences in transmission-constrained regions.

The facility class scope encompasses the main generating unit — boiler, steam turbine-generator, and associated balance-of-plant equipment — the coal handling and storage system from rail or barge unloading through the on-site coal pile to the pulverizers, the fly ash and bottom ash collection systems, the coal combustion residual (CCR) impoundment or dry disposal system, and the plant DCS and SCADA systems governing generation and environmental control operations. Natural gas pipeline supply to any coal units with auxiliary gas-firing capability is

addressed by reference to DBA-ES-GC-FC4 and DBA-GAS-FC1 without displacement of those instruments.

1.3 Relationship to Governing Documents

This document is issued under the DBA-ES Electric Sector Framework (v0.7), which is itself issued under the DBA-EN-SF Energy Sector Framework (v0.3) and the DBA-MA Parent Framework (IAEA L1 v1.4). The threat taxonomy, bounding methodology, consequence analysis structure, and adequacy standards are inherited from the sector framework. This facility-class document adds the coal-generation-specific physical configuration, asset taxonomy, bounding events, and consequence chains that the sector framework establishes in general terms.

The inheritance map for this document is as follows. The threat actor taxonomy (Tier 1, 2, 3), the four characteristics of military action (intentionality, persistence, precision, denial), the three-tier consequence framework, the GenCo/TransCo/DisCo/CntlCo terminology, and the adequacy standards are inherited unchanged from the DBA-ES sector framework. The Dual Design Basis architecture and Bright Line / Command Broker requirements are inherited unchanged and applied to the plant DCS and boiler management environment in Section 3.3. The RT_HVLLC metric — established in the DBA-ES series for transmission equipment — is applied here to thermal generation long-lead components.

1.4 The Dirt Burner as a Strategic Asset — Why Declining Does Not Mean Disposable

The coal fleet's decline is well-documented and ongoing. What is less well-documented is the grid services function of the surviving fleet. Large coal turbine-generators provide synchronous inertia — the kinetic energy stored in rotating masses that resists instantaneous frequency changes when generation and load become unbalanced. As coal retires and is replaced by inverter-based solar and wind resources, the grid's total synchronous inertia decreases. The surviving coal units disproportionately serve transmission-constrained load pockets where their reactive power output provides local voltage support that cannot be easily replicated by remote resources, regardless of the energy market economics.

An adversary who targets these specific units — not coal generation in general, but the units that the RTO has designated as must-run for voltage support in a specific transmission pocket — is targeting not the energy they produce but the grid services they provide. The energy can often be replaced; the reactive power support from a specific geographic location cannot be replaced without either a new local resource or a major transmission expansion. This is the precision characteristic of military action applied to grid topology knowledge: the adversary who knows which coal units are transmission-essential and attacks them specifically produces a consequence that is disproportionate to the nameplate capacity retired.

2. Facility Profile and Asset Taxonomy

2.1 Boiler and Steam Generation

Subcritical boilers. Operating at steam pressures below approximately 3,200 psi and temperatures below approximately 1,000°F. Subcritical boilers are the dominant technology in older coal plants and represent the majority of the remaining U.S. coal fleet by unit count. Their pressure parts — waterwall tubes, superheater and reheater tubing, and steam drums — are the primary long-lead recovery components. Replacement of major pressure part sections requires specialized boiler fabrication contractors and has lead times of 12 to 24 months for major repairs.

Supercritical and ultra-supercritical (USC) boilers. Operating above the critical point of water (approximately 3,200 psi), where the distinction between liquid and steam phases disappears. Supercritical and USC plants operate at higher efficiency than subcritical units and represent newer additions to the coal fleet. Their once-through boiler design — with no steam drum — uses different and more specialized pressure part materials (high-chromium alloys) whose procurement lead times can reach 24 to 36 months for major components, establishing the upper bound of the RT_HVLLC metric for this facility class.

Pulverizers. Coal pulverizers — bowl mills, ball-tube mills, or roller mills — grind raw coal to the fine powder required for combustion. A large coal unit typically has six to eight pulverizers, each capable of supplying approximately one-sixth to one-eighth of full load. Pulverizer destruction reduces unit output proportionally; destruction of all pulverizers eliminates output. Pulverizer replacement lead times are 6 to 12 months for major grinding elements, shorter than the boiler pressure parts but still governing for partial-capacity restoration scenarios.

2.2 Steam Turbine-Generator

The steam turbine-generator is the highest-consequence single piece of equipment at a coal plant from a recovery timeline perspective. Large steam turbines for coal generation are custom-engineered to each unit's steam conditions, speed, and output rating. There are very few manufacturers globally capable of producing large steam turbines for conventional coal plants. Replacement of a destroyed steam turbine — including procurement, manufacturing, transportation to the site, and installation — has a lead time of 18 to 36 months, establishing the governing RT_HVLLC metric for Scenario A. The generator rotor is similarly long-lead at 12 to 24 months. Unlike gas turbines, for which spare rotors can sometimes be obtained from the OEM inventory or from other operators, coal steam turbines are sufficiently plant-specific that no meaningful strategic spare program exists or is practically feasible at the national level.

2.3 Coal Handling and Storage

Rail unloading. The majority of U.S. coal plants receive coal by unit train — typically 100 to 135 cars carrying approximately 10,000 to 13,500 tons per delivery. The rail unloading facility

— rotary car dumper or bottom-dump hopper — is the coal plant's interface with its fuel supply chain. Destruction of the rail unloading facility eliminates the plant's primary fuel receipt capability; barge unloading (where available as an alternative) may sustain operations at reduced delivery rates.

On-site coal pile. Most large coal plants maintain 30 to 90 days of coal inventory in on-site storage piles. The coal pile is the plant's buffer against fuel supply disruption — a rail network outage that would shut down a plant with 7 days of inventory has no consequence for a plant with 60 days. The coal pile inventory level at the time of an attack determines how long the plant can continue to operate after fuel supply is interdicted. Pile size is constrained by available land area and coal degradation (oxidation and spontaneous combustion risk for high-volatile coals).

Conveyor and transfer systems. The internal coal handling system — conveyors, transfer towers, and crushers — moves coal from the pile to the pulverizers. Destruction of key transfer points in the internal handling system stops coal flow to the burners regardless of pile inventory, with a repair timeline of weeks rather than months.

2.4 Coal Combustion Residual Management

Coal combustion produces two primary solid waste streams: fly ash (fine particulate captured by precipitators or fabric filters) and bottom ash (coarser material collected from the furnace bottom). Both are classified as coal combustion residuals (CCRs) under EPA's 2015 CCR rule (40 CFR Part 257). Disposal methods include wet impoundments — large surface water retention structures that hold CCR in slurry form — and dry disposal in lined landfills. The CCR impoundment is the environmental consequence multiplier for FC6: its structural integrity is the governing parameter for the Scenario C compound consequence chain.

The Kingston Fossil Plant failure (2008) — where an impoundment dike failure released approximately 1.1 billion gallons of CCR slurry — demonstrated that a wet CCR impoundment failure produces an environmental consequence that is orders of magnitude larger than any other environmental release associated with a thermal power plant. The slurry covered approximately 300 acres, reached the Emory and Clinch Rivers, and contaminated drinking water sources across a wide area—full remediation required over a decade and approximately \$1.2 billion. The DBA-MA extension adds the adversarial pathway: a precision attack on the impoundment dike structure or the drainage infrastructure that deliberately produces the same consequence.

2.5 Plant DCS and Environmental Controls

The plant DCS governs boiler combustion control, steam temperature and pressure management, turbine control, and emissions monitoring. Environmental control systems — selective catalytic reduction (SCR) for NO_x, wet or dry flue gas desulfurization (FGD) for SO₂, and electrostatic precipitators or fabric filters for particulate — are increasingly integrated into the plant DCS. For coal plants operating under tight emissions permit conditions, loss of environmental controls can

trigger an emissions permit compliance issue that requires the plant to derate or shut down independent of any physical damage to the generation equipment. An adversary who compromises the environmental control DCS and produces a permit violation condition achieves a generation curtailment without any physical attack on the power island.

2.6 Regulatory Authority

NERC CIP. Coal plants meeting the BES definition are subject to NERC CIP cybersecurity standards. CIP-014 physical security requirements apply to coal plants that NERC identifies as critical transmission system facilities — primarily those whose loss would produce Tier 2 grid consequences. The must-run and transmission-essential designation that establishes FC6 applicability for sub-750 MW units is likely to trigger CIP-014 coverage.

FERC. FERC has jurisdiction over wholesale electricity markets and the reliability standards applicable to generating facilities that participate in those markets. FERC's role in coal plant security is primarily through the NERC reliability standard framework.

EPA CCR Rule. 40 CFR Part 257 establishes groundwater monitoring, structural stability assessment, and closure requirements for CCR impoundments and landfills. The structural stability assessment requirement — which requires annual inspections by licensed engineers and corrective action for identified deficiencies — provides the regulatory baseline for the CCR impoundment Design Basis. The DBA-MA adversarial scenario is not addressed in the CCR rule's structural stability framework.

TSA. TSA voluntary pipeline security guidelines apply to the gas supply infrastructure that serves coal plants with auxiliary gas-firing capability. TSA has not issued mandatory physical security requirements for coal generation facilities, creating the same regulatory gap identified across the DBA-OIL and DBA-GAS series.

3. Design Basis: Coal-Fired Generation Facility Class

3.1 Non-Adversarial Bounding Events

The non-adversarial Design Basis for the coal-fired generation facility class inherits the process safety event, infrastructure failure event, natural hazard event, and operational event categories from the DBA-ES Sector Framework. This document develops the facility-class-specific consequence chains for three governing non-adversarial bounding events.

Boiler tube failure and forced outage. The governing process safety bounding event is a major boiler tube failure producing a forced outage during peak demand conditions. Boiler tube failures — typically waterwall tube leaks, superheater tube failures, or economizer tube corrosion breaches — are the most common cause of unplanned coal plant outages. The bounding case involves a major pressure part failure requiring replacement of a significant tube section, with a repair timeline of 4 to 8 weeks, long enough to produce Tier 2 grid reliability consequences in a

transmission-constrained system during extended peak demand. The bounding event establishes the non-adversarial baseline against which the Scenario A military-action extension is built.

Coal supply disruption — weather and railroad network. Coal supply disruption due to extreme weather affecting the railroad network — winter storms that cause network-wide slowdowns, flooding that closes river barge navigation, or derailments that block key supply corridors. The 2021 Winter Storm Uri and the recurring pattern of Powder River Basin railroad network disruptions during severe winters provide the reference events. The bounding case involves a 30-day coal supply interruption beginning with on-site inventory at 45 days, producing fuel depletion and forced shutdown at approximately day 75 unless alternative supply arrangements are made.

CCR impoundment structural anomaly. Detection of a structural anomaly in a wet CCR impoundment — seepage, embankment cracking, or settlement — requires emergency drawdown and potential plant shutdown pending structural repair. The Kingston event provides the reference for the maximum consequence of an uncontrolled failure; the bounding non-adversarial event is the controlled response to a detected anomaly. The repair and regulatory authorization timeline — typically 6 to 18 months for a significant structural repair — can exceed the plant's outage tolerance and trigger a permanent decommissioning decision.

3.2 Design Basis Accident — Military Action: Bounding Event

The DBA-MA bounding event for the coal-fired generation facility class is a coordinated attack that simultaneously destroys the highest-RT_HVLLC components of the steam turbine-generator and boiler pressure parts, depletes the on-site coal inventory through SCADA-based fuel management manipulation in the weeks preceding the kinetic event, and breaches the CCR impoundment structure to produce the environmental cascade as a compound consequence — executed at a must-run or transmission-essential unit during a high-demand, transmission-constrained system condition when no replacement generation is available in the affected load pocket.

The bounding parameters, inherited from the DBA-ES Sector Framework threat taxonomy:

Adversary tier: Tier 1 state actor with plant DCS pre-positioning capability and precision kinetic delivery. Ukrainian theater experience demonstrates that thermal power plants are established military action targets; the DBA-MA bounding event applies the precision and intentionality characteristics to select the specific components whose destruction produces the maximum RT_HVLLC outcome.

Modality: Cyber-kinetic combined arms. DCS pre-positioning manipulates the coal pile management system to increase coal consumption rates and delay delivery scheduling — depleting on-site inventory over a 4 to 6 week horizon before the kinetic event. Kinetic action targets the high-pressure steam turbine rotor and the main boiler drum or primary superheater

headers — the components with the longest individual replacement lead times. The CCR impoundment dike foundation is a secondary kinetic target.

Target selection: Must-run or transmission-essential units designated by the applicable RTO/ISO, selected for their role in providing local voltage support in transmission-constrained load pockets. The adversary who targets these units is not attacking coal generation in general; they are attacking the grid services that cannot be replaced by importing generation from adjacent regions.

Timing: Peak summer demand during a heat wave, or peak winter demand during a cold snap, when the transmission-constrained load pocket is under maximum stress, and the RTO has already designated the affected unit as must-run. The environmental persistence characteristic means the adversary maintains DCS access for the full 4 to 6 week coal inventory depletion phase before executing the kinetic component.

Simultaneity: Coordinated attacks on the highest-RT_HVLLC units at two or more coal plants serving the same transmission-constrained load pocket within a 72-hour window. The load pocket's total must-run capacity is eliminated simultaneously, producing a Tier 2 consequence that exceeds what any single-unit attack could achieve.

3.3 Dual Design Basis Architecture

Consistent with the DBA-ES Sector Framework and DBA-EN-SF (§4.2), a coal-fired generating facility may simultaneously carry a sector-specific physical Design Basis established by this document and an AI Governance Design Basis established under the AI Governance series. The two design bases interact at the DBA level. The Bright Line and Command Broker architecture established in CB-Framework_Cross_Sector v0.1 governs the boundary between the two design bases.

Three coal-generation-specific AI governance interaction points are identified for the FC6 Design Basis:

Boiler combustion optimization and APC. APC systems that continuously adjust burner firing patterns, excess air levels, and steam temperature profiles to maximize efficiency and minimize emissions are ML systems operating below the Bright Line when their set point recommendations remain within the boiler's design operating envelope. They are formally testable and may operate autonomously within those bounds. The adversarial concern is manipulation of the combustion APC to drive boiler operating conditions toward the limits that produce accelerated tube wear or thermal fatigue — the same optimization-safety boundary identified in DBA-GAS-FC2 for processing plant APC. The AI Governance Design Basis must establish that combustion APC operating envelopes include explicit safety margin setbacks from the boiler's tube temperature and pressure limits, that APC envelope modifications require management of change review, and that the APC system has no pathway to override or suppress a boiler safety system trip regardless of the optimization recommendation.

Coal inventory management and fuel supply scheduling ML. ML systems that manage on-site coal inventory levels, schedule rail deliveries, and optimize pile management — including blending of coals from different sources — are below the Bright Line when performing deterministic scheduling functions within their trained envelopes. This is precisely the attack surface exploited in the DBA-MA bounding event's pre-positioning phase: a compromised coal inventory management ML system that systematically increases consumption rates and delays delivery scheduling depletes the on-site stockpile over weeks without triggering operational anomaly detection. The AI Governance Design Basis must establish that coal inventory management ML systems are subject to periodic inventory audit against independent physical measurements, that cumulative inventory deviations from seasonal delivery plans trigger human review rather than automatic adjustment, and that delivery schedule changes that would reduce pile inventory below a defined minimum buffer level require Command Broker authorization.

Environmental control DCS and emissions management Gen AI. Gen AI advisory tools for emissions compliance optimization — tools that recommend SCR reagent injection rates, FGD limestone feed rates, and precipitator operating parameters based on real-time fuel analysis and emissions permit conditions — are above the Bright Line and require Command Broker authorization. The coal plant context adds the emissions permit compliance dimension: a Gen AI tool that recommends reduced environmental control performance — reduced reagent injection, reduced sorbent feed — may produce near-term cost savings while driving the plant toward a permit violation that triggers mandatory curtailment. The Command Broker must evaluate not only the immediate operational reasonableness of the recommendation but its cumulative effect on permit compliance margins.

4. Design Basis Accident Scenarios

4.1 Scenario A: Precision Kinetic Attack on Boiler and Turbine — RT_HVLLC Governing

4.1.1 Initiating Event

Precision kinetic attack against a must-run coal unit serving a transmission-constrained load pocket, targeting the high-pressure steam turbine and the primary superheater headers of a supercritical boiler. The DCS pre-positioning phase, executed over the preceding 5 weeks, has reduced the on-site coal pile from its normal 60-day inventory to approximately 15 days by manipulating the coal inventory management system to increase pulverizer throughput slightly above design rate and delay the next scheduled unit train delivery. The coal pile depletion is within the range that a seasonal demand variation could explain; it has not triggered a procurement alert.

The kinetic component executes during a peak summer heat wave when the unit is at maximum load, and the RTO has issued a must-run instruction: precision munitions against the high-

pressure turbine casing and against the primary superheater header assembly at the boiler outlet. The turbine casing breach produces immediate rotor damage from blade-casing contact; the superheater header destruction eliminates the boiler's ability to produce steam at supercritical conditions. Both components are on the critical path of the RT_HVLLC calculation.

4.1.2 Progression Sequence

T+0: Turbine trip and emergency boiler shutdown initiated automatically by the plant protection system. Loss of generation — approximately 800 MW — instantaneous. RTO/ISO notified. Transmission operators activating emergency operating procedures for the affected load pocket: voltage support degraded, reactive reserve reduced.

T+0 to T+24 hours: Physical damage assessment: high-pressure turbine rotor destroyed, primary superheater headers severed. Coal pile inventory at 15 days. DCS forensic investigation initiated — the coal inventory management manipulation is identified, but attribution requires detailed forensic analysis. Emergency coal delivery ordered — first available unit train 5 days out.

T+24 hours to T+2 weeks: Turbine OEM contacted for replacement rotor assessment. OEM confirms no available spare; custom rotor manufacturing required. Lead time estimate: 22 to 30 months. Superheater header replacement estimate: 14 to 18 months for high-alloy supercritical material. RT_HVLLC governing component: turbine rotor at 22 to 30 months. Transmission operator implementing emergency congestion management for the affected load pocket; neighboring units operating above normal ratings.

T+2 weeks onward: Plant in extended forced outage. Turbine rotor on order; manufacturing initiated. RTO/ISO conducting emergency transmission planning to manage the load pocket without the must-run unit — options include transmission reinforcement, demand response contracts, new local generation procurement, and import capability enhancement. All are measured in years, not months.

4.1.3 Consequence Envelope

RT_HVLLC governing metric. The supercritical steam turbine rotor is the governing long-lead component: 22 to 30 months from procurement to installation and commissioning. This is the longest RT_HVLLC in the DBA-ES generation series, exceeding even the large power transformer timeline that governs DBA-ES-TX-FC2. The plant is effectively removed from service for the duration — no partial operation is possible without the turbine rotor.

Transmission consequence — load pocket. The loss of a must-run unit in a transmission-constrained load pocket produces a Tier 2 grid reliability consequence that the transmission operator cannot fully offset through normal emergency operating procedures. The load pocket must be managed through a combination of emergency ratings on neighboring circuits, demand response activation, and potentially rotating outages during peak demand periods — for the full

22 to 30 month outage duration. This is a sustained operational reliability degradation, not a transient event.

Synchronous inertia loss. The loss of a large coal steam turbine-generator removes a significant synchronous inertia contribution from the interconnection. In regions where coal retirements have already reduced synchronous inertia, the loss of an additional large unit increases the rate of change of frequency (ROCOF) following a subsequent generation trip, potentially triggering under-frequency load shedding at a threshold that would not have been reached with the coal unit in service.

4.2 Scenario B: Coal Supply Interdiction — Rail and Barge Disruption

4.2.1 Initiating Event

Coordinated attack against the railroad infrastructure serving the primary coal supply corridor for a cluster of coal plants in the Appalachian or Powder River Basin supply region — specifically targeting rail bridges, tunnel entrances, or switching yards at hydraulic chokepoints in the railroad network where alternative routing is unavailable. The attack is designed to produce a sustained coal supply disruption of 45 to 60 days, long enough to deplete the on-site coal inventories of plants operating near normal stock levels (30 to 45 days) but short enough that the attack does not require attacking the generating facilities themselves.

4.2.2 Fuel Depletion Without Physical Plant Damage

Scenario B is the coal fleet's analog of the pipeline voluntary shutdown problem identified in DBA-OIL-FC4: the generation equipment is physically undamaged, but the plant cannot operate because its fuel supply has been severed. The consequence is a forced shutdown that is indistinguishable in its grid effect from physical plant destruction — the generation is gone — but with a fundamentally different recovery profile. A coal plant forced to shut down due to fuel depletion will restart within days of coal supply restoration, without any equipment procurement or regulatory authorization delay.

The adversary who chooses Scenario B over Scenario A is making a strategic calculation: the fuel interdiction attack is operationally simpler, requires no precision knowledge of plant internal configuration, produces an equivalent grid consequence in the short term, and — critically — the target is the railroad infrastructure rather than the generating facility, which may have a lower security posture and less sophisticated physical protection. The rail bridge or tunnel that is the chokepoint in the coal supply network may be a far more accessible target than the power plant it feeds.

4.2.3 Consequence Envelope: Fuel Interdiction

Fleet-wide impact. A successful attack on a major coal supply corridor affects not one plant but every plant drawing coal from that corridor. The Powder River Basin supplies approximately 40

percent of U.S. coal-fired generation. A sustained PRB rail corridor disruption — affecting the BNSF and UP main lines through the Wyoming coal corridor — forces fuel depletion shutdowns across a significant portion of the remaining U.S. coal fleet simultaneously. The grid consequence is a multi-state, multi-region capacity shortage that cannot be quickly offset by any single alternative supply or generation source.

Recovery timeline. Coal supply restoration following railroad bridge or tunnel repair is governed by civil engineering timelines rather than power plant equipment procurement. A major railroad bridge repair: 4 to 12 weeks, depending on the structure and the extent of damage. The coal plant restarts within days of supply restoration, assuming the on-site pile can be rebuilt to minimum operating levels — typically 7 to 10 days of inventory. Total restoration timeline: 5 to 14 weeks from attack to full generation restoration, significantly shorter than Scenario A but longer than any non-coal generation fuel supply disruption.

Cross-modal fuel shifting. Coal plants with rail supply disruptions may attempt to shift to barge delivery where waterway access is available, or to emergency truck delivery for small supplemental volumes. These alternatives are capacity-limited and significantly more expensive but can extend the operational runway beyond the on-site pile inventory. The adversary who understands the modal alternatives will target the rail chokepoint during a period when waterway navigation is also restricted — ice or low water — to eliminate the backup option simultaneously.

4.3 Scenario C: Combined Attack with CCR Impoundment Breach — Generation Loss Plus Environmental Cascade

4.3.1 Initiating Event

Compound attack combining a Scenario A precision kinetic strike on the generating equipment with a simultaneous structural attack on the coal combustion residual wet impoundment at the same facility — specifically targeting the downstream dike section where foundation conditions are most vulnerable. The CCR impoundment at a large coal plant typically holds tens to hundreds of millions of gallons of ash slurry; its failure produces an uncontrolled release comparable in scale to the Kingston event.

4.3.2 The Compound Consequence Chain

The CCR impoundment breach adds a second, independent consequence chain to the Scenario A generation loss. The generation consequence — grid reliability impact, RT_HVLLC recovery timeline — proceeds on the same timeline as Scenario A. The environmental consequence chain operates on a separate, parallel timeline that is governed not by equipment procurement but by remediation scope, regulatory jurisdiction, and downstream water system impact.

The impoundment slurry release reaches the nearest waterway within hours of the dike breach. The downstream consequence depends on the receiving water body: a release to a small tributary

that feeds a municipal water supply reservoir produces a drinking water contamination consequence; a release to a navigable waterway produces a navigation closure and fisheries consequence. Both are Tier 3 cross-sector cascades — the environmental consequence reaches the water sector, consistent with the DBA-EN-SF cross-sector elevation rule.

4.3.3 Consequence Envelope: Compound Attack

Environmental remediation timeline. Full remediation of a major CCR slurry release to a waterway is measured in years, as demonstrated by the Kingston event (approximately 11 years from spill to EPA closure of the site remediation). The active remediation period — during which the affected waterway and surrounding land are under remediation activities — restricts navigation, fishing, and recreational use of the affected waterway for the full duration. Downstream water utilities must activate alternative supply sources or treatment upgrades to address CCR constituents — arsenic, selenium, mercury — in source water.

Regulatory consequence cascade. A major CCR impoundment failure triggers EPA enforcement action, potential facility permit revocation, and state environmental regulatory response simultaneously. The regulatory response may affect not only the damaged facility but other coal plants with similar wet impoundment configurations in the same region — a regulatory cascade that amplifies the supply consequence beyond the directly attacked facility.

Cross-sector water system impact. The Tier 3 cross-sector cascade — CCR constituents reaching drinking water sources — connects the DBA-ES consequence chain to the DBA-W series. A CCR slurry release that contaminates a major drinking water source in the same watershed as the affected plant activates the water sector emergency response framework simultaneously with the electric sector response, creating compound federal coordination demands that exceed the capacity of any single federal agency.

5. Protective Action Thresholds

5.1 Threat Detection and Indicator Taxonomy

5.1.1 Elevated Threat Indicators

Intelligence Community reporting of adversary interest in U.S. thermal generation infrastructure or specific coal plant control system architectures. Anomalous plant DCS network activity: unusual access to coal inventory management systems, anomalous combustion APC parameter queries, and lateral movement between corporate IT and plant control networks inconsistent with normal maintenance operations. Systematic reduction in on-site coal pile inventory below seasonal norms without a corresponding demand explanation. Physical surveillance of plant perimeters, railroad corridor chokepoints, or CCR impoundment access roads. Unusual aerial or ground activity near CCR impoundment structures. Procurement inquiries for steam turbine specifications or boiler pressure part material specifications for specific plant configurations.

5.1.2 Imminent Threat Indicators

Confirmed unauthorized plant DCS access. Coal pile inventory approaching minimum operating threshold without a credible demand or supply explanation. Confirmed unauthorized aerial UAS in plant airspace or over CCR impoundment structures. Simultaneous anomalous DCS events at two or more coal plants within the same transmission-constrained load pocket. Confirmed kinetic event at any coal plant or railroad infrastructure serving the coal supply corridor: anomalous seepage or structural movement at a CCR impoundment inconsistent with normal seasonal variation.

5.2 Decision Authority Framework

Plant operator level: Plant shift supervisor has the authority to initiate emergency shutdown, isolate DCS systems, and notify the grid operator of an unexpected outage. NERC CIP incident reporting for confirmed or suspected cyber incidents affecting plant control systems. EPA notification for any CCR impoundment structural anomaly or release. PHMSA notification is not applicable to coal plants absent auxiliary gas facilities triggering DBA-ES-GC-FC4 coordination. Local emergency management notification for CCR release events with off-site consequence potential.

Grid operator level: RTO/ISO emergency operating procedures for loss of must-run generation: emergency ratings, demand response activation, and import maximization. NERC event analysis notification. FERC notification for significant reliability events.

Federal coordination: CISA for plant DCS cyber incident coordination. DOE for supply disruption affecting regional reliability. EPA for CCR impoundment failures with environmental consequences. FBI and DOD for events with confirmed or strongly suspected state-actor attribution. Army Corps of Engineers for navigation closure coordination following CCR waterway releases.

5.3 Federal Coordination Thresholds

Confirmed kinetic attack on any coal plant meeting the FC6 applicability criteria. Confirmed plant DCS compromise with suspected nation-state attribution. Loss of must-run or transmission-essential generation for more than 72 hours in a transmission-constrained load pocket without an alternative mitigation plan in effect. CCR impoundment release reaching or threatening to reach a navigable waterway or public water supply. Simultaneous generation loss at two or more coal plants serving the same transmission-constrained load pocket within 72 hours. Coal supply corridor disruption is affecting fuel deliveries to more than one plant for more than 30 days.

6. Recovery Architecture

6.1 RT_HVLLC as the Governing Recovery Metric

The RT_HVLLC metric — Recovery Timeline for High-Value Long-Lead Components — was established in the DBA-ES series for transmission equipment (large power transformers and converter transformers) whose destruction produces the longest recovery timelines in the electric sector. FC6 extends this metric to thermal generation, where the supercritical steam turbine rotor establishes the governing RT_HVLLC of 22 to 30 months — longer than the LPT RT_HVLLC of 18 to 24 months established in DBA-ES-TX-FC2.

The RT_HVLLC metric governs Scenario A recovery planning and drives the primary recommendation in §7: the only recovery action that can meaningfully reduce the Scenario A RT_HVLLC is pre-positioned availability of critical long-lead components — turbine rotors and primary superheater headers for the specific supercritical plant configurations that represent the highest-consequence attack targets. No regulatory, operational, or emergency response action taken after the attack can compress the manufacturing timeline for a custom steam turbine rotor.

6.2 Recovery Sequencing

Phase 1 — Immediate (0–72 hours). Emergency plant shutdown and damage assessment. Grid operator emergency operating procedures for loss of must-run generation. Plant DCS forensic investigation initiation. Coal supply situation assessment — pile inventory level and days-to-depletion. Steam turbine OEM notification and initial damage assessment. CCR impoundment structural assessment for Scenario C events. EPA notification for any CCR release.

Phase 2 — Short-term (72 hours to 8 weeks). Steam turbine OEM damage assessment complete; replacement rotor procurement initiated. Boiler pressure part damage assessment and repair scope determination. DCS forensic investigation continuing. Grid operator long-term congestion management planning for an extended must-run outage. Coal supply restoration (Scenario B) — railroad repair and alternative supply arrangements—environmental remediation initiation for Scenario C CCR events.

Phase 3 — Medium-term (8 weeks to RT_HVLLC completion). Steam turbine rotor manufacturing (22 to 30 months). Boiler pressure part procurement and repair (14 to 18 months, running in parallel). DCS reconstitution and re-commissioning. Sequential return to service following all component replacement and NERC documentation updates—ongoing CCR environmental remediation for Scenario C events, independent of generation return-to-service timeline.

6.3 Critical Recovery Constraints

Steam turbine rotor manufacturing. The governing RT_HVLLC constraint for Scenario A: 22 to 30 months from procurement authorization to installed and commissioned replacement rotor. No strategic spare rotor program exists or is practically feasible for the diverse specifications of the remaining U.S. coal fleet. The only actionable mitigation is the long-term grid planning

recommendation in §7.2: reducing transmission-constrained load pocket dependency on must-run coal units before the attack occurs, rather than attempting to accelerate the recovery after.

Supercritical boiler pressure part procurement. High-chromium alloy pressure part materials for supercritical and USC boilers have a global supply chain concentrated in a small number of specialized manufacturers. Lead times of 14 to 18 months for major header sections can extend further if the adversary has pre-interdicted the relevant manufacturing supply chain. No strategic reserve exists.

DCS reconstitution and plant restart authorization. Plant DCS forensic investigation and reconstitution: 4 to 6 weeks minimum, consistent with other DBA-ES instruments. The coal plant context adds the requirement that the coal inventory management ML system be re-validated against its delivery scheduling baseline before automated inventory management can resume — the reconstituted system must be re-certified to operate within the inventory management bounds established by the Design Basis.

CCR remediation — independent timeline. CCR environmental remediation runs on a timeline independent of the generation return-to-service timeline. The plant can be returned to generation service when the generation equipment is repaired, and the regulatory path to restart is clear; the CCR remediation continues for years thereafter. The remediation timeline does not gate generation restart but imposes ongoing legal, financial, and reputational consequences that affect the plant owner's willingness and ability to invest in the generation recovery.

7. Gaps and Recommendations

7.1 Regulatory Gap: Absence of Mandatory Physical Security Standards for Coal Generation

Coal plants meeting the FC6 applicability criteria — must-run and transmission-essential units — are subject to NERC CIP cybersecurity standards but have no mandatory physical security standard equivalent to what CIP-014 provides for transmission substations. CIP-014 requires vulnerability assessments and physical security plans for transmission facilities whose loss would produce significant reliability consequences; an analogous requirement for must-run and transmission-essential generating facilities does not exist. NERC, working with FERC, should evaluate extending the CIP-014 physical security framework to generating facilities designated as must-run or transmission-essential, using the FC6 DBA-MA consequence envelope as the design basis for the vulnerability assessment scope.

7.2 Transmission Planning for Must-Run Dependency Reduction

The most consequential long-term mitigation for the FC6 Scenario A consequence is reducing the transmission system's dependency on must-run coal generation for voltage support in constrained load pockets — through transmission reinforcement, new local synchronous

condensers, or advanced inverter-based reactive power resources that can provide the voltage support currently requiring synchronous generation. FERC and the RTOs should require transmission planning studies that specifically assess the grid's resilience to the loss of must-run coal units in each transmission-constrained load pocket, using the FC6 RT_HVLLC of 22 to 30 months as the planning horizon. The planning obligation should be linked to the must-run designation itself. If a unit is designated must-run, the RTO should have a documented plan for operating the affected transmission pocket without that unit for a period equal to or exceeding the RT_HVLLC.

7.3 Coal Supply Corridor Vulnerability Assessment

The railroad infrastructure serving the major coal supply corridors — particularly the Powder River Basin BNSF and UP main lines — has not been assessed against the DBA-MA threat tier. The coal supply chokepoint concept identified in Scenario B — railroad bridges and tunnel entrances whose destruction would interrupt coal supply to multiple generating facilities simultaneously — is not addressed in any existing federal vulnerability assessment framework. DOE, in coordination with TSA and the Surface Transportation Board, should commission a coal supply corridor vulnerability assessment that identifies the highest-consequence chokepoints and evaluates hardening options, alternative routing, and supply stockpiling requirements at affected plants.

7.4 CCR Impoundment Structural Security Assessment

EPA's CCR rule requires structural stability assessments of wet impoundments but does not require security assessments against adversarial structural attack. The DBA-MA Scenario C consequence chain — CCR impoundment breach producing an environmental cascade to the water sector — is not addressed in any existing regulatory framework. EPA, in coordination with the Army Corps of Engineers, should develop a CCR impoundment security assessment requirement — analogous to the dam safety security assessment framework under FERC Part 12 — that addresses the adversarial structural attack pathway identified in this document.

7.5 RT_HVLLC Component Strategic Awareness

No federal agency maintains visibility into the procurement lead times and global manufacturing availability for supercritical steam turbine rotors and high-alloy boiler pressure parts for the specific plant configurations that represent the highest-consequence FC6 attack targets. DOE's Office of Electricity maintains the large power transformer reserve program for transmission equipment; an analogous awareness function — tracking the manufacturing queue, global supplier capacity, and lead time data for critical thermal generation long-lead components — would provide the federal government with the information needed to assess the RT_HVLLC consequence of specific attack scenarios and to evaluate whether pre-positioned procurement agreements are feasible for the highest-consequence units.

Revision History

Version	Date	Author	Description
0.1	June 2026	T. Roxey	Initial version. Establishes Design Basis Accident — Military Action scenarios for the coal-fired generation facility class. Issued under DBA-ES Electric Sector Framework v0.7. FC designation DBA-ES-GN-FC6 established; GN type code introduced for the GenCo generation sub-series. Applicability threshold: individual unit nameplate capacity at or above 750 MW, or RTO/ISO must-run or transmission-essential designation. Three scenarios were developed: Scenario A (precision kinetic attack on boiler and turbine — RT_HVLLC of 22 to 30 months for supercritical turbine rotor as governing metric), Scenario B (coal supply interdiction via railroad infrastructure attack — fuel depletion without physical plant damage), Scenario C (combined attack with CCR impoundment breach — generation loss plus water sector environmental cascade). Dual Design Basis architecture established with three coal-generation-specific AI governance interaction points. RT_HVLLC metric extended from transmission equipment to thermal generation. Regulatory gaps identified: mandatory physical security for must-run generation, transmission planning for must-run dependency reduction, coal supply corridor vulnerability assessment, CCR impoundment structural security assessment, and RT_HVLLC component strategic awareness.

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